

Pluralistic Alignment via Group-Relative Policy Optimization

with a Graph-Grounded Coverage Reward

PluralTree

1 Setup and notation

Let q be an open-ended question on which people genuinely disagree. The knowledge graph $\mathcal{G}$ is a rooted tree over survey data (ROOT $\rightarrow$ topic $\rightarrow$ subtopic $\rightarrow$ question $\rightarrow$ axis $\rightarrow$ opinion), where an *axis* is a (question $\times$ demographic attribute) pair and each opinion leaf carries one subgroup’s probability distribution over that question’s answer options.

A retrieval module (the *divergence scout*) selects an anchor $a(q) \in \mathcal{G}$ and renders its subtree into a context block, giving the policy input x_q . Retrieval is executed once, at dataset-construction time: the graph encoder and the scout are **frozen** throughout this phase, so what is trained is the reasoner’s use of retrieved evidence, not the retrieval itself.

2 Graph-grounded positions

For anchor $a(q)$ we aggregate the opinion leaves of its subtree into a set of population-level *positions*

$$\mathcal{P}(q) = \{(p_1, \pi_1), \dots, (p_n, \pi_n)\}, \quad \pi_k = \frac{1}{|S_k|} \sum_{g \in S_k} \Pr_g(\text{option } k),$$

where S_k is the set of subgroups answering the underlying survey item and π_k is the mean subgroup probability of option k (its *prevalence*). Positions below a prevalence floor are discarded as noise.

This is the crux of the design: the reward is defined against *measured human survey distributions*, not against a judge model. It therefore (i) introduces no dependence on the held-out evaluation benchmark, and (ii) can be computed on any question with a graph anchor, giving effectively unlimited training signal.

3 Reward

Let y be a generated answer, segmented into position-units $u_1, \dots, u_m$ (enumerated items or sentences). Let $e(\cdot)$ be a *held-out* sentence embedder — deliberately distinct from the encoder the scout retrieves with, so the reward cannot credit merely retrieval-shaped text — and write $s_{jk} = \langle e(u_j), e(p_k) \rangle$ for unit-normalised embeddings.

Mention. Position k is *mentioned* if some unit matches it:

$$\text{MEN}_k(y) = \mathbf{1} \left[\max_j s_{jk} \geq \tau \right].$$

Depth. Each unit is attributed to its single best-matching position, $\alpha(j) = \arg \max_k s_{jk}$, and contributes its length only if that match clears τ :

$$\text{dep}_k(y) = \sum_{j: \alpha(j)=k, s_{j\alpha(j)} \geq \tau} |u_j|,$$

with $|u_j|$ the word count of unit j . Note MEN and dep use different assignment rules: a position may be *mentioned* by a unit whose best match lies elsewhere.

Coverage. A position counts only when it is both named and articulated,

$$\text{COV}_k(y) = \text{MEN}_k(y) \wedge \mathbf{1}[\text{dep}_k(y) \geq d], \quad (1)$$

for a depth threshold d . Equation (1) is the central correction relative to a mention-only reward. Empirically, an answer that names every position in a single clause each scores near zero on the target metric while a mention-only reward scores it as full recall; without the depth term the reward is blind to the dominant failure mode.

Terms. With τ the match threshold and T an effective position count,

$$R(y) = \frac{1}{n} \sum_{k=1}^n \text{COV}_k(y) \quad (\text{recall, uniform over positions}) \quad (2)$$

$$P(y) = \frac{1}{m} \sum_{j=1}^m \mathbf{1} \left[\max_k s_{jk} \geq \tau \right] \quad (\text{precision; penalises padding}) \quad (3)$$

$$V(y) = \frac{\max(0, \sum_k \text{COV}_k(y) - T)}{T}, \quad T = \exp(H(\pi)) \quad (\text{over-enumeration}) \quad (4)$$

Recall is *unweighted* even though prevalences π_k are available. The evaluation metric counts every viewpoint cluster equally; weighting recall by prevalence would train the policy to serve the majority and neglect exactly the minority positions pluralism exists to represent.

Composition. The terms combine multiplicatively,

$$r(y) = R(y) \cdot P(y)^{\lambda_p} \cdot \frac{1}{1 + \lambda_v V(y)} \in [0, 1], \quad (5)$$

rather than as subtracted penalties. Additive penalties admit a compensatory effect — the policy inflates one term to pay for another — whereas factors in $[0, 1]$ cannot be traded off, and (5) needs no clamping. We set $\lambda_v = 0$ by default: the evaluation metric is monotone in the number of clusters covered, so it cannot reward commitment, and T is derived from *graph* prevalence skew, which does not imply few *human* viewpoint clusters. The anti-enumeration role is instead served by the depth term in (1): listing many positions shallowly leaves none above d .

4 Objective

For each prompt x_q we sample a group of G completions $\{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot \mid x_q)$ and score each with (5). The group supplies its own baseline, so no value network is required:

$$A_i = \frac{r(y_i) - \mu}{\sigma + \varepsilon}, \quad \mu = \frac{1}{G} \sum_i r(y_i), \quad \sigma = \text{std}(r(y_1), \dots, r(y_G)). \quad (6)$$

Writing $\rho_{i,t}(\theta) = \frac{\pi_{\theta}(y_{i,t} \mid x_q, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t} \mid x_q, y_{i,<t})}$, the objective is the clipped policy-gradient surrogate with a KL anchor to the reference policy:

$$\mathcal{J}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(\rho_{i,t} A_i, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) A_i \right) \right] - \beta \mathbb{D}_{\text{KL}}[\pi_{\theta} \parallel \pi_{\text{ref}}]. \quad (7)$$

A consequence of (6) worth stating: if every rollout in a group receives the same reward, $\sigma \rightarrow 0$ and the group contributes no gradient. This is benign for the per-response reward (5), whose value depends on response content, but it becomes a real hazard for any *group-dependent* reward (e.g. crediting each rollout by its marginal contribution to the group’s union coverage), where perfect complementarity and total collapse both yield symmetric rewards and hence a vanishing advantage.

5 Policy

π_{θ} is a LoRA adapter over a trainable bf16 instruction-tuned base ($\sim 7\text{B}$); the evaluation-time reasoner is a larger quantised model and is not trainable. Consequently any comparison must include a *same-base* no-training control: a policy may improve over its own base and still sit below the larger model’s untrained baseline.

6 Leakage protocol

The evaluation benchmark’s questions and its human ratings are held out absolutely. Training prompts are drawn from the graph’s own survey questions, and any question whose normalised text matches a benchmark item is dropped at dataset-construction time. The reward (5) never consults human ratings or a judge model, so no evaluation signal can enter training through the reward path.

7 Hyperparameters

Symbol	Meaning	Value
G	rollouts per prompt (group size)	8
β	KL coefficient to π_{ref}	0.04
ϵ	PPO clip range	0.2
τ	match threshold (cosine)	0.50
d	depth threshold (words/position)	60 [†]
λ_p	precision exponent	0.5
λ_v	over-enumeration scale	0
–	sampling temperature	1.0
–	LoRA rank / α	16 / 32
–	learning rate	10 ^{−6}

Table 1: [†] Provisional. d should be calibrated by regressing the probability that a viewpoint cluster is satisfied on the number of words the response devotes to it, using the benchmark’s own human-rated reference responses — a fit that requires no judge model.

8 Group-diversity objective (proposed; gated)

The largest measured effect in this project is across samples: union coverage over separately generated answers is 0.66–0.69, against about 0.51 for any single answer. The objective below rewards each rollout for viewpoints the rest of its group did not express. It uses no targets and no judge, so it applies to any training question, and it replaces (5) in (6) unchanged.

Pool. For a group $\{y_i\}_{i=1}^G$, all units of all rollouts are greedily clustered by cosine similarity at threshold τ_p into a pool $\mathcal{K}$. A unit joins its best cluster if the match clears τ_p , otherwise it opens a new one. No minimum support is imposed: a viewpoint voiced by a single rollout is kept.

Covered set. Rollout i articulates cluster k if some unit matches it and at least d_p words of its units are attributed to k , by the same mention and depth rules as (1). Write $C_i \subseteq \mathcal{K}$.

Reward.

$$q_i = \frac{|C_i|}{|\mathcal{K}|}, \quad n_i = \frac{|C_i \setminus \bigcup_{j \neq i} C_j|}{|\mathcal{K}|}, \quad r_i = q_i (1 + \lambda n_i). \quad (8)$$

n_i is the rollout’s leave-one-out contribution to the group’s union; $\lambda = 0$ recovers within-answer coverage of the pool. In a *reference* variant, n_i instead removes clusters covered by samples of the frozen base model for the same prompt, which credits a viewpoint that several rollouts discover together. Neither variant can separate identical rollouts, so a fully collapsed group still gives zero advantage.

Gate. Training is conditional on a within-question test against the judge. For rollouts i of one question, with C_i^J the clusters the judge marks as covered, the judge target is $t_i = |C_i^J \setminus \bigcup_{k \neq i} C_k^J| + |C_i^J|$. Let κ be the pairwise concordance of r with t , and κ_J the concordance of the judge with itself when re-run on a different participant subset. Variants $(\lambda, d_p, \text{mode})$ are chosen on a random half of the questions; on the other half the reward passes iff $\kappa_J > 0.5$, $\kappa \geq 0.5 + 0.5(\kappa_J - 0.5)$, the bootstrap lower bound of κ exceeds 0.5, and concordance with $|C_i^J|$ alone is at least 0.5.

To log per group (not yet implemented). Reward standard deviation, the fraction of zero-variance groups, $|\mathcal{K}|$, and the mean novelty $\bar{n}$, which separate a group that has collapsed from one whose reward carries no signal.

Open issues

1. **Reward–metric agreement has been measured, and fails.** The advantage (6) depends only on the *within-group ordering* of rewards, so the relevant statistic is within-question pairwise concordance with the evaluation judge. It is 0.152 at the defaults ($\tau = 0.50$, $d = 60$) and at best 0.301 across a sweep of $\tau \in [0.25, 0.50]$ and $d \in [0, 150]$, against chance 0.500. The failure has two regimes and no setting escapes both: at low τ few pairs tie but the reward orders them wrongly (concordance among separated pairs 0.417), while at high τ 73% of pairs tie. Recalibrating τ or d therefore cannot repair it, and training on (5) as specified is not justified. The judge itself is not validated at this level (aggregate $\rho = 0.167$ against the benchmark authors’ 0.88, ceiling unmeasured), so the concordance target is noisy too.

2. **Reachability.** (7) reweights the reference policy’s existing distribution; it cannot create behaviour the base model never samples. A best-of- K sweep on the frozen base establishes whether the target behaviour is reachable at all.
3. **Construct gap — the leading explanation for item 1.** Positions are survey answer options aggregated over demographic subgroups, about 19.9 per question. The evaluation’s viewpoint clusters, about 7.75 per question, are formed by k -means over participants’ agree/disagree votes on one another’s written statements. The two differ in both kind and granularity, so a response can cover many positions while covering few clusters. Re-targeting the reward at cluster-shaped objectives is the direct test and has not yet completed.
4. **Calibration of d .** See table footnote; the reward’s absolute scale is not meaningful until d is fit.